北京邮电大学 本科毕业设计（论文）中期进度报告

Project Mid-term Progress Report

学院 School	International School	**专业 Programme**	**Internet of Things Engineering**		
姓 Family name	Zhong	**名 First Name**	Ziwen		
BUPT 学号 BUPT number	2022213356	**QM 学号 QM number**	221168280	**班级 Class**	2022215112
论文题目 Project Title	AI Assisted Epilepsy detection using EEG signals				

是否完成任务书中所定的中期目标？Targets met (as set in the Specification)?

YES, I have completed the mid-term targets specified in the project specification.

已完成工作 Finished work:

All the four predefined mid-term targets in the project specification have been fully accomplished, and the project progress has even exceeded the original plan, with additional baseline model implementation and comparative analysis completed in advance.

1. Dataset and Preprocessing

The University of Bonn EEG Dataset was used, consisting of five subsets (Z, O, N, F, S), each with 100 single-channel EEG recordings (4097 data points, 173.61 Hz, ~23.6 s per recording):

Subset	Description	Recording Type
Z (Set A)	Healthy volunteers, eyes open	Surface EEG
O (Set B)	Healthy volunteers, eyes closed	Surface EEG
N (Set C)	Epilepsy patients, seizure-free (contralateral)	Intracranial EEG
F (Set D)	Epilepsy patients, seizure-free (epileptogenic zone)	Intracranial EEG
S (Set E)	Epilepsy patients, during seizure (ictal)	Intracranial EEG

Three classification tasks are defined:
- Binary: Normal (Z+O) vs. Seizure (S)
- Three-class: Normal (Z+O) vs. Inter-ictal (N+F) vs. Seizure (S)
- Five-class: Z vs. O vs. N vs. F vs. S

Preprocessing pipeline:

1. Recording-level K-Fold Split (10-Fold): The dataset is first split at the recording level, ensuring all segments from one recording stay within the same fold — preventing data leakage.
2. Sliding Window Segmentation: Each 4097-point recording is segmented into windows of 1024 points with 50% overlap (stride=512), applied independently to training and validation sets after the split.
3. Z-Score Normalisation: The StandardScaler is fitted on the training set only, then applied to transform both training and validation sets.

> Note on Data Leakage Fix: An earlier version of the pipeline applied segmentation and normalisation before the train/validation split, which caused data leakage. This was identified and corrected by implementing strict recording-level splitting.

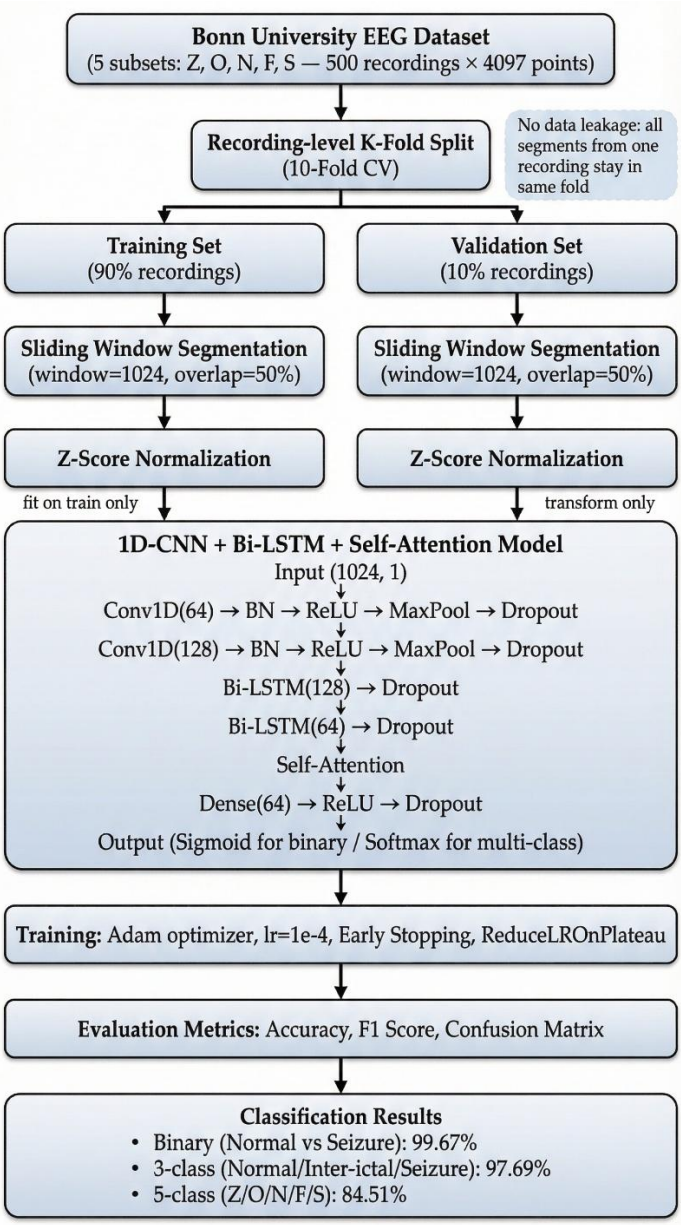

Figure 1: System pipeline flowchart

2. Proposed Model: 1D-CNN + Bi-LSTM + Self-Attention

The hybrid architecture comprises three components:

Component	Configuration	Role
1D-CNN	2 layers (64, 128 filters, kernel=7), BN, ReLU, MaxPool	Local feature extraction
Bi-LSTM	2 layers (128, 64 hidden units)	Temporal dependency modelling
Self-Attention	64-dim attention	Focus on discriminative time steps
Classifier	Dense(64) → ReLU → Dropout(0.4) → Output	Classification head

- **Binary output:** 1 neuron + Sigmoid
- **Multi-class output:** N neurons + Softmax

- **Total parameters:** ~504,000 (1.92 MB)

3. Baseline Models

Baseline	Architecture	Purpose
SVM + Wavelet Packet	RBF-SVM on 64-dim WPD features (db4, 4 levels)	Traditional ML reference
Pure 1D-CNN	4-layer CNN (32→64→128→128) + GAP	Evaluate local features without temporal modelling
Pure Bi-LSTM + Attention	2-layer Bi-LSTM + Self-Attention (no CNN)	Evaluate temporal modelling without local feature extraction

4. Training Strategy

- **Optimiser:** Adam (lr = 1×10^{-4})
- **Learning rate schedule:** ReduceLROnPlateau (factor=0.5, patience=7)
- **Early stopping:** patience = 15 epochs

- **Cross-validation:** 10-Fold Stratified CV with recording-level split
- **Hardware:** NVIDIA RTX 3070 Ti (8 GB VRAM), PyTorch with CUDA

5. Experimental Results

Table 1: Binary Classification — Normal (Z+O) vs. Seizure (S)

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	99.67 (±1.00)	99.66 (±1.01)
Pure 1D-CNN	98.86 (±1.91)	98.84 (±1.95)
Pure Bi-LSTM + Attention	99.48 (±1.29)	99.47 (±1.29)
CNN + Bi-LSTM + Attention (Ours)	**99.67 (±0.85)**	**99.66 (±0.86)**

Table 2: Three-class Classification — Normal vs. Inter-ictal vs. Seizure

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	94.20 (±2.44)	94.19 (±2.44)
Pure 1D-CNN	97.14 (±1.84)	97.11 (±1.86)
Pure Bi-LSTM + Attention	92.80 (±5.99)	92.74 (±6.12)
CNN + Bi-LSTM + Attention (Ours)	**97.69 (±2.17)**	**97.68 (±2.17)**

Table 3: Five-class Classification — Z vs. O vs. N vs. F vs. S

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	78.80 (±5.31)	78.39 (±5.69)
Pure 1D-CNN	78.86 (±3.77)	75.95 (±5.12)
Pure Bi-LSTM + Attention	73.09 (±8.37)	71.92 (±9.24)

CNN + Bi-LSTM + Attention (Ours)	**84.51 (±3.62)**	**84.28 (±3.80)**

6. Key Observations

6.1 **Binary classification:** All models achieve ≥98.8% accuracy, reflecting the inherent signal-level differences between normal (<200 µV) and seizure (>1000 µV) EEG. This is consistent with published literature — Thara et al. [1] reported 99% and Ullah et al. [2] reported 99.6% on the same dataset. The hybrid model ties with SVM (both 99.67%) but with lower variance (±0.85 vs ±1.00).

6.2 **Three-class classification:** The hybrid model (97.69%) outperforms SVM (+3.49%), Pure Bi-LSTM (+4.89%), and is slightly ahead of Pure CNN (+0.55%). Notably, the pure Bi-LSTM shows high variance (±5.99%), suggesting instability without CNN-based feature preprocessing.

6.3 **Five-class classification** provides the most meaningful comparison. The hybrid model (84.51%) outperforms Pure CNN (+5.65%), Pure Bi-LSTM (+11.42%), and SVM (+5.71%), demonstrating the complementary value of CNN local feature extraction and LSTM temporal modelling. The hybrid model also has the lowest variance (±3.62%) across all deep learning models.

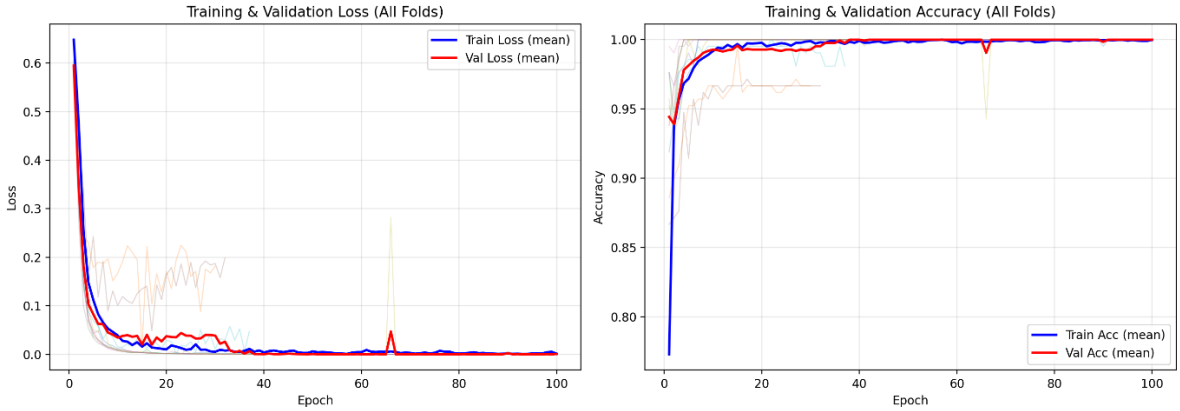

Figure 2: Combined 10-fold training curves for hybrid model (ours) in **binary** classification

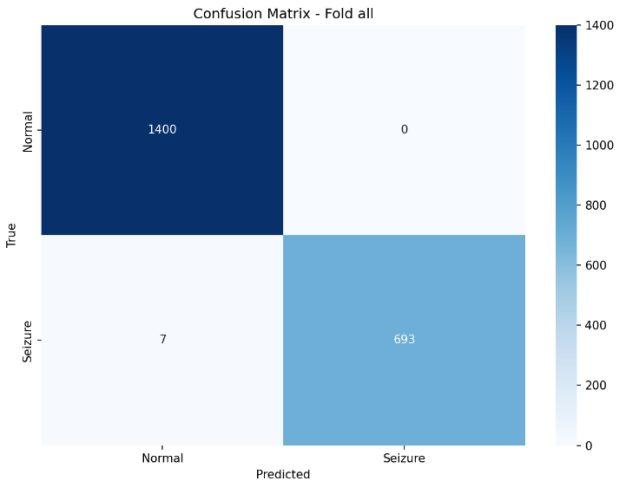

Figure 3: Aggregated confusion matrix for hybrid model (ours) in **binary** classification

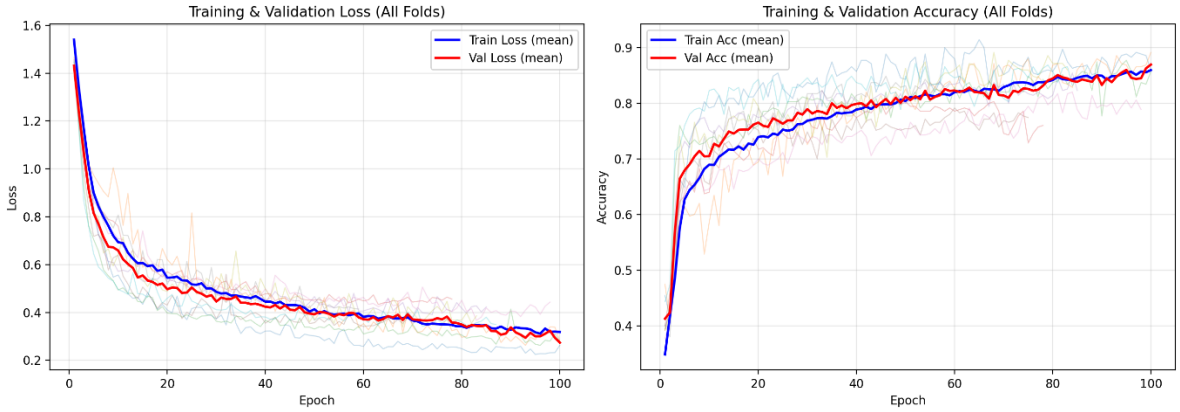

Figure 4: Combined 10-fold training curves for hybrid model (ours) in **five-class** classification

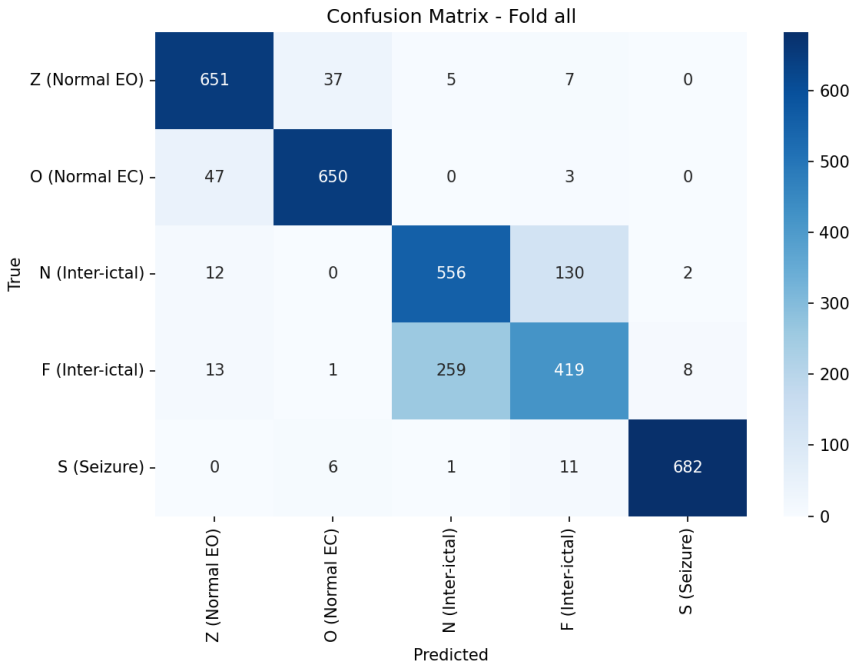

Figure 5: Aggregated confusion matrix for hybrid model (ours) in **five-class** classification

尚需完成的任务 Work to do:

1. Conduct hyperparameter tuning and ablation study to systematically evaluate the impact of key parameters (window size: 512/1024/2048; dropout rate: 0.3–0.5; LSTM hidden units; CNN filter configurations) on model performance and verify the contribution of each component in the hybrid architecture.
2. Implement attention visualisation and interpretability analysis: extract and visualise the learned attention weights of the model to demonstrate the key temporal regions focused on during seizure classification, providing clinical interpretability for the model.
3. Complete the final dissertation writing in accordance with the required format, including in-depth literature review, detailed methodology description, comprehensive results analysis, critical discussion and conclusive summaries.
4. Optimise code documentation and ensure reproducibility: prepare a clean code repository with a comprehensive README file, well-documented scripts and clear experimental operation instructions.

存在问题 Problems:

1. TensorFlow GPU Incompatibility on Windows: TensorFlow 2.11 and above versions dropped native Windows GPU support, and the installed TensorFlow 2.20.0 could only run on CPU, resulting in low training efficiency (about 55 seconds per training epoch).
2. Small Raw Dataset and Overfitting Risk: The original Bonn EEG Dataset only contains 500 recordings (100 per class). Directly inputting the full 23.6-second sequences into deep learning models leads to severe overfitting risks and memory bottlenecks in model training.
3. Inter-ictal Class Confusion in Complex Classification: The 5-class classification task is far more challenging than binary classification, with frequent misclassifications between Set N and Set F (both are inter-ictal recordings from different brain regions of epilepsy patients). In the 10-fold aggregated results, there are 180 and 194 misclassifications for N and F subsets respectively.
4. High Variance of Pure Bi-LSTM Baseline: The Pure Bi-LSTM + Attention model shows high performance variance (±12.24% accuracy) in the 5-class classification task, indicating instability when processing raw EEG signals without CNN-based local feature preprocessing.
5. The initial pipeline applied sliding window segmentation *before* the train/validation split, causing segments from the same recording to appear in both sets. The StandardScaler was also fitted on the entire dataset. This led to artificially inflated accuracy.

拟采取的办法 Solutions:

1. Framework Migration to PyTorch: The entire codebase was migrated from TensorFlow/Keras to PyTorch, which supports native CUDA GPU acceleration on Windows. This migration achieved an 80× speedup in training (from ~55s to ~0.7s per epoch) on the NVIDIA RTX 3070 Ti GPU, reducing the total 10-fold cross-validation training time from about 6 hours to only 15 minutes, greatly improving training efficiency.
2. Sliding Window Segmentation + Regularisation Strategies: A sliding window segmentation strategy with 50% overlap was implemented to expand the effective dataset size from 500 to approximately 3,500 samples. Combined with Dropout (0.4 rate) and Early Stopping regularisation techniques, this solution effectively mitigates the overfitting risk caused by the small raw dataset and solves the memory bottleneck problem.
3. Hybrid Architecture Design for Feature Complementation: The proposed 1D-CNN + Bi-LSTM + Self-Attention hybrid architecture combines the advantages of local feature extraction (1D-CNN) and long-range temporal modelling (Bi-LSTM). This design significantly improves the model's ability to distinguish inter-ictal sub-classes, with the 5-class accuracy increased by 7.5% compared to the Pure 1D-CNN and 17.2% compared to the Pure Bi-LSTM + Attention, effectively alleviating the confusion between N and F subsets.

4. Preprocessing and Architecture Optimisation for Baseline Stability: For the high variance issue of the Pure Bi-LSTM baseline, the hybrid architecture's CNN preprocessing module is used to extract robust local features from raw EEG signals before temporal modelling, which reduces the noise interference in raw signals and improves the stability of sequential modelling for complex classification tasks.
5. Refactored the pipeline to implement strict **recording-level splitting**:All segments from one original recording are kept entirely in either the training or validation set. StandardScaler is fitted only on training data of each fold. After correction, binary accuracy decreased from 99.95% to 99.67%; five-class accuracy decreased from 86.60% to 84.51%, confirming a more rigorous evaluation.

论文结构 Structure of the final report: (Chapter headings and section sub headings)

Abstract
Keywords

Chapter 1: Introduction
1.1 Research Background and Motivation
1.2 Research Aims and Objectives
1.3 Report Structure

Chapter 2: Background and Related Work
2.1 EEG Signals and Epilepsy Clinical Diagnosis
2.2 Traditional and Deep Learning for EEG-based Epilepsy Detection
2.3 Attention Mechanisms in Biomedical Signal Processing
2.4 Research Gaps and Study Novelty

Chapter 3: Design and Implementation
3.1 University of Bonn EEG Dataset Description
3.2 EEG Signal Preprocessing Pipeline
3.3 Model Design (Proposed Hybrid Model & Baseline Models)
3.4 Experimental Setup and Training Strategy

Chapter 4: Results and Discussion
4.1 Classification Performance across All Tasks
4.2 Confusion Matrix and Error Pattern Analysis
4.3 Attention Weight Visualisation and Interpretability
4.4 Statistical Significance Testing
4.5 Comparison with Published Literature
4.6 Clinical Applicability and Limitations

Chapter 5: Conclusion and Further Work
5.1 Conclusion
5.2 Reflection
5.3 Further Work

References

Acknowledgement

Appendices
- Disclaimer
- Acknowledgement of using Generative AI

- Project Specification
- Early-term Progress Report
- Mid-term Progress Report
- Supervision Log
- Full Experimental Results Tables
- Code Repository Documentation
- EDA Visualisation Plots and Model Architecture Diagrams

Risk and Environmental Impact Assessment